Dyadic Lucas Prime Cascades and Composite Fibonacci Shifts

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Abstract

Let P be a positive odd integer, and let $U_0 = 0$, $U_1 = 1$, $U_{n+2} = PU_{n+1} + U_n$. We prove that, for every integer h, the set of $r \geq 1$ for which $U_{2^r} + h$ is composite has positive lower natural density. The proof gives a more precise propagation statement: a sufficiently late prime value divides an arithmetic progression of later values. An explicit threshold is obtained from the two quantities $v_2((P^2+4)(h-1)^2+3)$ and $v_2((P^2+4)(h+1)^2+3)$. The argument combines exact convergence of the even and odd dyadic subsequences in $\mathbb{Z}_2$ with return of finite-field powers after their 2-primary orders have been exhausted. Taking $P = 1$ answers Saito's Problem 1.8 for every integer shift, with positive lower density in place of infinitude. All sequence identities and valuation calculations used in the proof are established here. The source identifiers T43 and T44 are retained only for cross-reference with the author's research notes; historical priority beyond the cited literature is not claimed.

1 The problem and the main theorem

Saito's Problem 1.8 in the April 2025 version of [1] asks whether, for each fixed integer h, the Fibonacci values

$$F_{2^r} + h \qquad (r \geq 1) \tag{1}$$

are composite infinitely often. Here $F_0 = 0$, $F_1 = 1$ and $F_{n+2} = F_{n+1} + F_n$. The question keeps both the Fibonacci sequence and the dyadic index 2^r fixed. We keep these quantifiers and answer it through a general Lucas sequence result.

For $S \subseteq \mathbb{N}$, write

$$\underline{d}(S) = \liminf_{X\to\infty} \frac{\#(S \cap [1, X])}{\lfloor X \rfloor}.$$

Throughout, a composite number is a positive integer with a proper positive divisor other than 1. Thus negative and zero values, or the value 1, are not called composite; each can occur only finitely often in the increasing sequences considered below. For nonzero rational numbers, v_2 denotes the exponent of 2, normalized by $v_2(2) = 1$. It extends to $\mathbb{Q}_2$ in the usual way, with $v_2(0) = +\infty$.

Fix a positive odd integer P. Define

$$U_0 = 0, \quad U_1 = 1, \quad U_{n+2} = PU_{n+1} + U_n, \qquad D = P^2 + 4. \tag{2}$$

For an integer h, put

$$B_{P,h} = \max\left\{v_2\big(D(h-1)^2+3\big), v_2\big(D(h+1)^2+3\big)\right\}. \tag{3}$$

The two arguments of v_2 are positive integers, so this threshold is always finite and can be computed directly from P and h.

Theorem 1.1 (Prime cascade; [source T43]). *Suppose that $r \geq B_{P,h} + 1$ and*

$$p = U_{2^r} + h$$

is a prime satisfying $p \nmid 2D$. There exists an integer $k \geq 1$ such that

$$p \mid U_{2^{r+jk}} + h \qquad \text{for every } j \geq 1. \tag{4}$$

Every number in (4) is larger than p and is composite. Consequently,

$$\underline{d}\{r \geq 1 : U_{2^r} + h \text{ is composite}\} > 0 \tag{5}$$

for every positive odd P and every $h \in \mathbb{Z}$.

The step k is effectively obtainable once a prime seed satisfying the hypotheses is supplied: it is a multiplicative order in a finite cyclic group. The density assertion does not assert that a prime seed exists. If there are no sufficiently late prime values, all sufficiently late values are already composite. This alternative is essential to the proof and removes any need to prove the existence of primes in the sequence.

Corollary 1.2 (Fibonacci shifts; [source T44]). *For every integer h,*

$$\underline{d}\{r \geq 1 : F_{2^r} + h \text{ is composite}\} > 0.$$

In particular, the assertion posed in Saito's Problem 1.8 is true for all integer shifts.

The main theorem has three ingredients. First, the dyadic Lucas values have two opposite 2-adic limits. Second, those limits are roots of a quadratic equation with no rational integer solutions; hence their 2-adic distances from the two relevant integer shifts have bounded valuations. Third, these bounds make the remaining finite-field root order odd. Repeated squaring then becomes invertible and returns to the initial state. Sections 2–5 give the complete argument; Sections 6–8 record examples, certification procedures and the exact scope.

2 The dyadic Lucas state

Let $V_0 = 2$, $V_1 = P$, and $V_{n+2} = PV_{n+1} + V_n$ be the companion Lucas sequence. Set

$$f_r = U_{2^r}, \qquad \ell_r = V_{2^r} \qquad (r \geq 1).$$

These two integer coordinates suffice for every part of the proof.

Lemma 2.1 (Doubling identities). *For all $r \geq 1$,*

$$f_{r+1} = f_r \ell_r, \qquad \ell_{r+1} = \ell_r^2 - 2, \tag{6}$$

$$\ell_r^2 - D f_r^2 = 4. \tag{7}$$

Moreover, $f_1 = P$, $\ell_1 = P^2 + 2$, every f_r is odd, and $\ell_r \equiv 3 \pmod 4$.

Proof. Let α, β be the roots of $X^2 - PX - 1$. Then $\alpha + \beta = P$, $\alpha\beta = -1$ and $(\alpha - \beta)^2 = D$. Direct substitution in the recurrence and its initial conditions gives

$$U_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}, \qquad V_n = \alpha^n + \beta^n.$$

The identities

$$U_{2n} = U_n V_n, \qquad V_{2n} = V_n^2 - 2(-1)^n, \qquad V_n^2 - DU_n^2 = 4(-1)^n$$

follow by expansion. Taking the even integer $n = 2^r$ gives (6)–(7). Since P is odd, $\ell_1 = P^2 + 2 \equiv 3 \pmod 8$ and f_1 is odd. If $\ell_r \equiv 3 \pmod 4$, then $\ell_r^2 - 2 \equiv 3 \pmod 4$; (6) also preserves oddness of f_r. □

The real sizes cause no difficulty. The sequence U_n is positive for $n \geq 1$ and strictly increasing for $n \geq 2$. This follows directly from its recurrence; the only initial equality is $U_1 = U_2 = 1$ when $P = 1$. In particular, f_r is strictly increasing for $r \geq 1$ and $f_r \to \infty$. These facts will turn divisibility into proper divisibility after a prime seed.

3 Two exact 2-adic limits

The real growth of f_r coexists with 2-adic convergence on its two parity classes. We need the exact rates, not merely existence of the limits.

Lemma 3.1 (Valuation identities)**.** *For every $r \geq 1$,*

$$v_2(\ell_r + 1) = r + 1, \tag{8}$$
$$v_2(f_{r+2} - f_r) = r + 1. \tag{9}$$

Proof. Since $P^2 \equiv 1 \pmod 8$, one has $v_2(\ell_1 + 1) = v_2(P^2 + 3) = 2$. Also

$$\ell_{r+1} + 1 = (\ell_r - 1)(\ell_r + 1).$$

The congruence $\ell_r \equiv 3 \pmod 4$ gives $v_2(\ell_r - 1) = 1$, proving (8) by induction. For the second identity, (6) gives

$$\begin{aligned} f_{r+2} - f_r &= f_r(\ell_r \ell_{r+1} - 1) \\ &= f_r(\ell_r^3 - 2\ell_r - 1) \\ &= f_r(\ell_r + 1)(\ell_r^2 - \ell_r - 1). \end{aligned}$$

The first and last factors are odd. Equation (8) therefore proves (9). □

Lemma 3.2 (Parity limits)**.** *There exist odd elements $u_0, u_1 \in \mathbb{Z}_2$, with $u_1 = -u_0$, such that $f_r \to u_i$ through $r \equiv i \pmod 2$. They satisfy*

$$Du_i^2 = -3 \tag{10}$$

and the exact approximation formula

$$v_2(f_r - u_{r \bmod 2}) = r + 1. \tag{11}$$

Neither u_0 nor u_1 is a rational integer.

Proof. For either fixed parity, (9) shows that the successive differences have valuations tending to infinity. The corresponding subsequence is Cauchy in the complete ring $\mathbb{Z}_2$. Its limit is odd because all terms are odd. In the convergent series

$$u_{r \bmod 2} - f_r = \sum_{j \geq 0} (f_{r+2j+2} - f_{r+2j}),$$

the first summand has valuation $r+1$ and all remaining summands have valuation at least $r+3$. The ultrametric inequality, with a unique term of least valuation, gives (11). Equation (8) says $\ell_r \to -1$. Passing to the limit in $f_{r+1} = f_r \ell_r$ along either parity gives $u_1 = -u_0$. Passing to the limit in (7) gives $1 - Du_i^2 = 4$, which is (10). If $u_i = c \in \mathbb{Z}$, this equality would hold in $\mathbb{Q}$ and hence in $\mathbb{R}$, contradicting $D > 0$ and $Dc^2 \geq 0$. □

No choice of a real square root of $-3/D$ is involved. The roots in Lemma 3.2 are constructed in $\mathbb{Z}_2$ by the recurrence. Their failure to be ordinary integers is precisely what provides a uniform distance bound for each fixed shift.

Lemma 3.3 (Distance from an integer)**.** *Let $u \in \{u_0, u_1\}$ and $c \in \mathbb{Z}$. Then*

$$v_2(u - c) + v_2(u + c) = v_2(Dc^2 + 3). \tag{12}$$

In particular, both valuations on the left are finite, nonnegative, and at most the quantity on the right.

Proof. The product identity

$$(u - c)(u + c) = u^2 - c^2 = -\frac{Dc^2 + 3}{D}$$

and the oddness of D give (12). Neither factor is zero by Lemma 3.2; both belong to $\mathbb{Z}_2$. □

4 From a shifted prime to an odd root order

The relevant finite-field root order divides either $p - 1$ or $2(p + 1)$. The next proposition explains why their 2-parts become bounded when p itself is a sufficiently late shifted sequence value.

Proposition 4.1 (The order budget)**.** *Let $r \geq B_{P,h} + 1$ and suppose $p = f_r + h$ is an odd prime. Then*

$$v_2(p - 1) \leq B_{P,h}, \qquad v_2(p + 1) \leq B_{P,h}. \tag{13}$$

Proof. Put $u = u_{r \bmod 2}$, $c_- = 1 - h$ and $c_+ = -1 - h$. Then $p - 1 = f_r - c_-$ and $p + 1 = f_r - c_+$. Lemma 3.3 gives

$$v_2(u - c_\pm) \leq B_{P,h}.$$

On the other hand, (11) gives $v_2(f_r - u) = r + 1 > B_{P,h}$. Thus, in $f_r - c_\pm = (f_r - u) + (u - c_\pm)$, the second summand has strictly smaller valuation. The valuation of the sum equals that of this summand, proving (13). □

Lemma 4.2 (Finite-field root orders)**.** *Let $p \nmid 2D$ be prime. A root γ of $X^2 - PX - 1$ in $\mathbb{F}_{p^2}$ has multiplicative order t satisfying*

$$\begin{cases} t \mid p - 1, & D \text{ is a square modulo } p, \\ t \mid 2(p + 1), & D \text{ is a nonsquare modulo } p. \end{cases}$$

If $p = f_r + h$ and $r \geq B_{P,h} + 1$, the order of $z = \gamma^{2^r}$ is odd.

Proof. The constant term is nonzero modulo p, so $\gamma \neq 0$. If D is a square modulo p, the roots lie in $\mathbb{F}_p^\times$, whose order is $p-1$. If D is a nonsquare, the Frobenius automorphism interchanges the two distinct roots. Since their product is -1,

$$\gamma^p = -\gamma^{-1}, \qquad \gamma^{p+1} = -1.$$

Thus $t \mid 2(p+1)$. Proposition 4.1 now implies $v_2(t) \leq B_{P,h} + 1 \leq r$. The order of γ^{2^r} equals $t/\gcd(t, 2^r)$ and has no factor 2. □

5 Return and positive lower density

We can now complete both the propagation statement and its density consequence without using any distribution theorem for prime values of Lucas sequences.

Proof of Theorem 1.1. Assume first the prime-seed hypotheses. Choose a root γ as in Lemma 4.2, put $z = \gamma^{2^r}$, and let q be its odd multiplicative order. If $q = 1$, set $k = 1$; otherwise set $k = \mathrm{ord}_q(2)$. In either case, for every $j \geq 1$,

$$z^{2^{jk}} = z, \qquad \text{hence} \qquad \gamma^{2^{r+jk}} = \gamma^{2^r}.$$

The other root is $\delta = -\gamma^{-1}$. All exponents under consideration are even, so the same equality holds with δ in place of γ. The finite-field version of Binet's formula is valid because $p \nmid D$:

$$U_n = \frac{\gamma^n - \delta^n}{\gamma - \delta} \quad \text{in } \mathbb{F}_{p^2}.$$

It follows that $U_{2^{r+jk}} \equiv U_{2^r} \pmod{p}$, proving (4). Strict increase gives $U_{2^{r+jk}} + h > U_{2^r} + h = p$, so the divisor is proper.

For the density conclusion, choose R so large that all $f_r + h$, $r \geq R$, exceed 1, exceed every prime divisor of $2D$, and satisfy $r \geq B_{P,h} + 1$. This is possible because $f_r \to \infty$. If no $f_r + h$ with $r \geq R$ is prime, every such value is composite and the composite-index set has density 1. Otherwise choose one prime seed $p = f_r + h$, $r \geq R$. The progression $r + k, r + 2k, \ldots$ from (4) lies in the composite-index set and has natural density $1/k$. Therefore the latter set has lower natural density at least $1/k > 0$. □

Proof of Corollary 1.2. The recurrence (2) with $P = 1$ is exactly the Fibonacci recurrence. Thus Theorem 1.1 applies with $D = 5$ and

$$B_{1,h} = \max\{v_2(5(h-1)^2 + 3), v_2(5(h+1)^2 + 3)\}.$$

It treats every fixed $h \in \mathbb{Z}$ and leaves the exponent 2^r unchanged. Positive lower density implies infinitude. □

Remark 5.1 (The parity shortcut). When h is odd, every $f_r + h$ is even and eventually greater than 2, so the composite-index set is cofinite. The finite-field argument is needed for even shifts. Keeping both parities in the theorem makes the quantifier over all integers h explicit.

6 Worked exact examples

6.1 A seed beyond the explicit threshold

Take $P = 1$, $h = -16$. Then

$$5(h-1)^2 + 3 = 1448 = 8 \cdot 181, \qquad 5(h+1)^2 + 3 = 1128 = 8 \cdot 141,$$

so $B_{1,-16} = 3$. At $r = 4$,

$$F_{16} - 16 = 987 - 16 = 971.$$

The integer 971 is prime: trial division by the primes at most $\sqrt{971} < 32$ suffices. Modulo 971, the roots of $X^2 - X - 1$ are 174 and 798. The first has order 485, and

$$174^{16} \equiv 481 \pmod{971}, \qquad \mathrm{ord}_{971}(481) = 485, \qquad \mathrm{ord}_{485}(2) = 48.$$

These equalities can be checked by modular powering, using $485 = 5 \cdot 97$ and $48 = 2^4 \cdot 3$ to test the proper prime-factor quotients of each claimed order. Therefore

$$971 \mid F_{2^{4+48j}} - 16 \qquad (j \geq 1). \tag{14}$$

This is an explicit infinite arithmetic progression of composite indices and gives the concrete lower bound $1/48$ for this shift. The bound need not be optimal.

6.2 A short return below the sufficient threshold

For $h = 2$, one has $B_{1,2} = 4$. The values

$$F_8 + 2 = 23, \qquad F_{16} + 2 = 989 = 23 \cdot 43$$

exhibit divisibility before the sufficient threshold $r \geq 5$. This does not contradict the theorem; the threshold is sufficient, not necessary. There is also a direct infinite certificate. At $r = 4$, the dyadic state modulo 23 is

$$(f_4, \ell_4) \equiv (21, -1) \pmod{23}.$$

The second coordinate remains -1 under $\ell \mapsto \ell^2 - 2$, while the first changes sign at each step. Hence $f_r \equiv 21 \pmod{23}$ for every even $r \geq 4$, and

$$23 \mid F_{2^r} + 2 \qquad (r \geq 4 \text{ even}).$$

The composite-index set for $h = 2$ therefore has lower density at least $1/2$ by this separate finite-state certificate.

6.3 Why the late-seed hypothesis cannot simply be dropped

Take $P = 1$, $h = 4$, $r = 2$. The value $F_4 + 4 = 7$ is prime, but it never divides a later shifted dyadic Fibonacci value. Indeed,

$$(f_2, \ell_2) = (3, 7) \equiv (3, 0) \pmod{7}.$$

The next first coordinate is zero and stays zero under $f \mapsto f\ell$. Thus

$$F_{2^s} + 4 \equiv 4 \pmod{7} \qquad (s \geq 3).$$

This disproves the unqualified assertion that every prime seed propagates. Here $B_{1,4} = 7$, so $r = 2$ is well outside the theorem's hypotheses. A proof that suppresses the order budget would miss this example.

7 A finite certificate for a supplied seed

Once P, h, r, p are given with $p = U_{2^r} + h$, the infinite divisibility statement admits a finite mathematical certificate. The following steps describe that certificate, not a search for arbitrary late prime values.

1. Certify that p is prime and that $p \nmid 2(P^2+4)$.
2. Evaluate (3) and verify $r \geq B_{P,h} + 1$.
3. Construct $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$ according as the discriminant is split or nonsplit, and choose a root γ of $X^2 - PX - 1$.
4. Compute $z = \gamma^{2^r}$, its odd order q, and a positive integer k with $2^k \equiv 1 \pmod{q}$; the exact order of 2 is optional if a larger valid k is easier to certify.
5. The identities in Section 5 certify every future index $r + jk$, so no further integer factorization is needed.

The root computation can also be replaced by a quotient-ring calculation modulo p. In either representation, a certificate is verified by finitely many exact arithmetic operations. A successful finite certificate proves a whole progression; checking a large number of isolated terms would not provide the same conclusion.

Proposition 7.1 (A convenient universal return step)**.** *Under the prime-seed hypotheses, let*

$$W = \frac{2(p^2-1)}{2^{v_2(2(p^2-1))}}.$$

If $W = 1$ set $k = 1$; otherwise any positive multiple of $\varphi(W)$ may be used in (4).

Proof. The root order divides $2(p^2-1)$ in either splitting case. After raising to 2^r, its remaining odd order q divides W. Since W is odd, Euler's theorem gives $2^{\varphi(W)} \equiv 1 \pmod{W}$, hence modulo q. The proof of Theorem 1.1 now applies. □

The step supplied by this proposition can be much larger than the least return step. Its purpose is to show that the certificate does not require a minimal-order calculation.

8 Scope, limitations and provenance

The proved assertion is about composite shifted values, not about the frequency of prime values. It allows the possibility that only finitely many shifted values are prime, and it does not prove that the composite-index set has a natural density. Lower density is the precise conclusion supported by the argument.

The restriction to odd P supplies the odd dyadic state and the exact valuation law (8). No assertion for all even P is made by this proof. Likewise, the recurrence has second characteristic coefficient -1 in the conventional Lucas notation $U_n(P, -1)$. Changing that coefficient changes the Pell identity and the limiting equation, so a separate argument is required.

The proof does not settle the infinitude of composite Fermat numbers. Its decisive input is a shift-dependent bound for the 2-primary parts of $p-1$ and $p+1$. For Fermat-type candidates those parts may grow with the exponent, and the odd-order conclusion need not hold. Nor does the Fibonacci specialization settle Saito's distinct cubic Pisot problem involving $\lfloor \beta^{3^r} \rfloor$, or determine the transcendence of Mills' constant.

The general theorem corresponds to T43 in the source research notes; Corollary 1.2 corresponds to T44. The doubling formulas, finite-field root orders and valuation rules are classical algebraic tools, proved explicitly above to make this manuscript independent. The contribution presented here is their assembled prime-seed propagation statement and its application to the exact published question. This draft does not identify a theorem as historically first merely because a literature search did not locate an identical result. Independent expert review and a fuller priority search remain appropriate before a priority claim is attached to publication.

References

[1] K. Saito, *Intervals without primes near an iterated linear recurrence sequence*, arXiv:2504.14968, version 1, 21 April 2025, Problem 1.8. arXiv:2504.14968. Published in *International Journal of Number Theory* (2026), doi:10.1142/S1793042126500685.